Quantum Computing Technology Roadmaps and Capability Assessment for Scientific Computing

An analysis of use cases from the NERSC workload

NERSC

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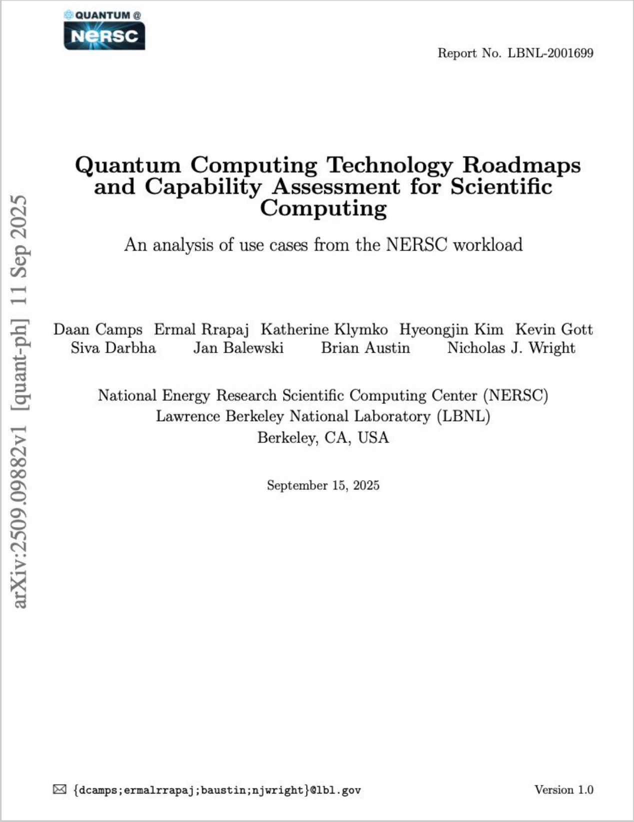
QUANTUM @ NERSC

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Quantum Computing Technology Roadmaps and Capability Assessment for Scientific Computing

An analysis of use cases from the NERSC workload

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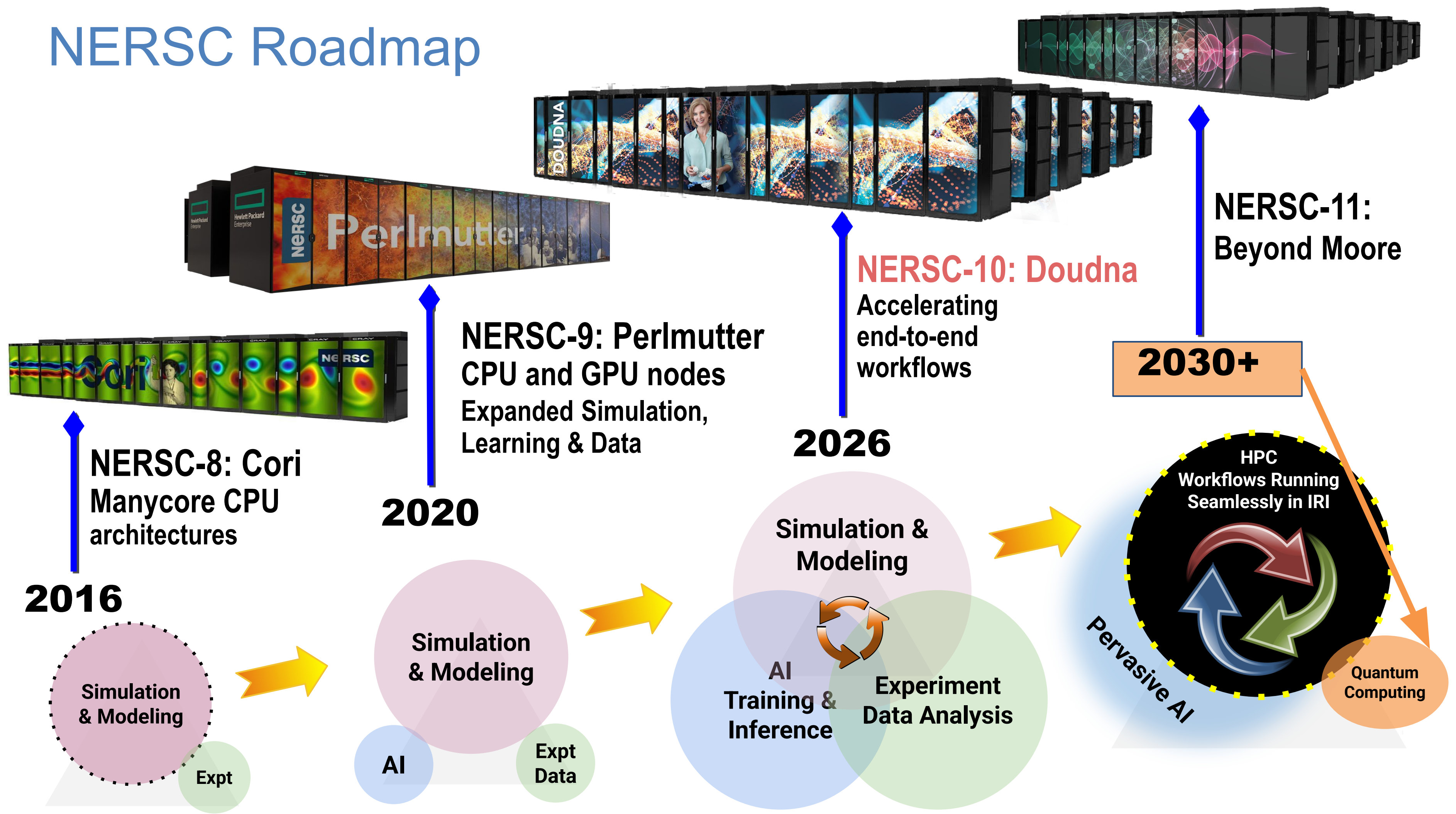
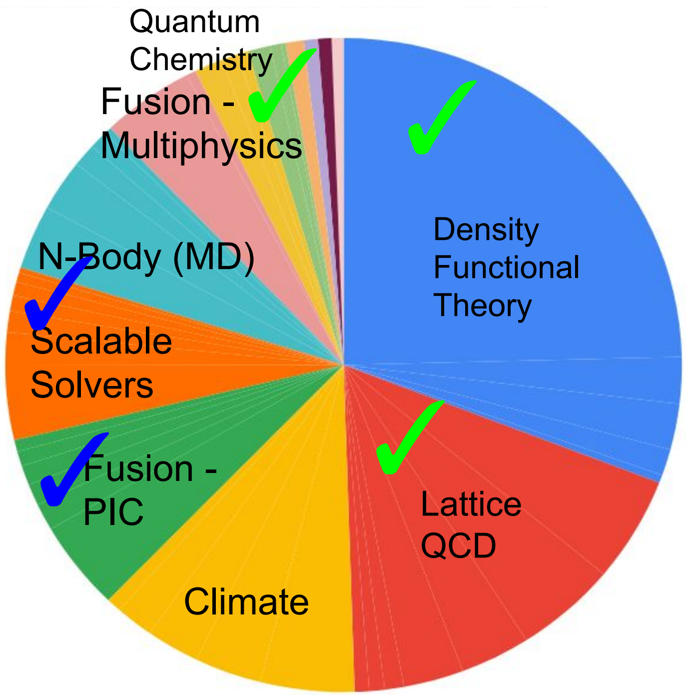
Version 1.0

White paper highlights

- Quantum computing is making **rapid progress**: algorithms becoming more efficient, computers becoming better.
- **Within 5 years**: projected overlap between capabilities of quantum computers & resource requirements for solving hard quantum physics problems.
- **Major risk**: orders of magnitudes improvements predicted by vendors – **Key**: transition to error corrected computers.
- In the 5-10 year time, the **speed of the QC** is likely to impact the feasibility of many applications.
- Proposed simple metric, **Sustained Quantum System Performance (SQSP)** to compare system level performance & throughput for a heterogeneous workload.


NERSC Roadmap
NERSC-8: Cori
Manycore CPU architectures
2016
NERSC-9: Perlmutter
CPU and GPU nodes
Expanded Simulation, Learning & Data
2020
NERSC-10: Doudna
Accelerating end-to-end workflows
2026
NERSC-11: Beyond Moore
2030+
Simulation & Modeling
Expt
Simulation & Modeling
AI
Expt Data
Simulation & Modeling
AI Training & Inference
Experiment Data Analysis
HPC Workflows Running Seamlessly in IRI
Pervasive AI
Quantum Computing

NERSC Workload Analysis Drives Quantum Computing Requirements



✓ Quantum mechanical problem
Quantum algorithms proposed

>50% of cycles

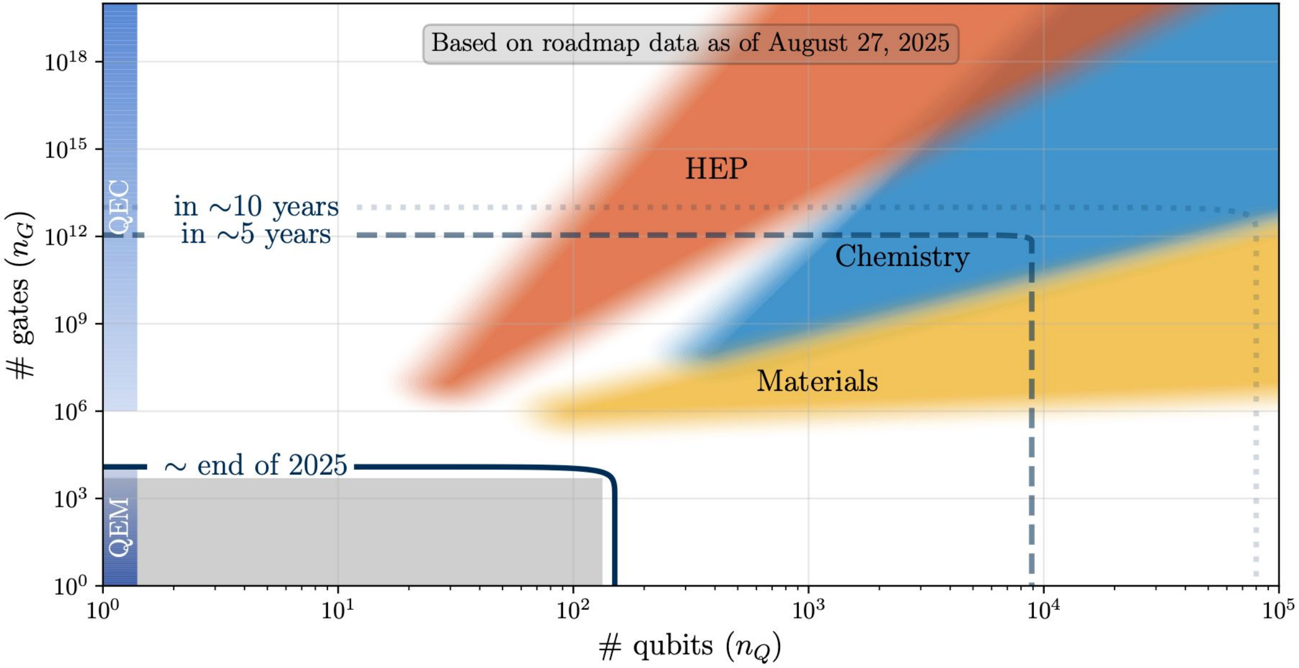
→ *materials science, chemistry, HEP*

✓ non-Quantum mechanical problem
Quantum algorithms proposed

20% of cycles

What is **not** on the pie chart?
Growth areas for quantum technologies!

Is a five to ten year time frame possible?



wedges = application requirements of science domains

lines = capabilities of QC based on vendor predictions

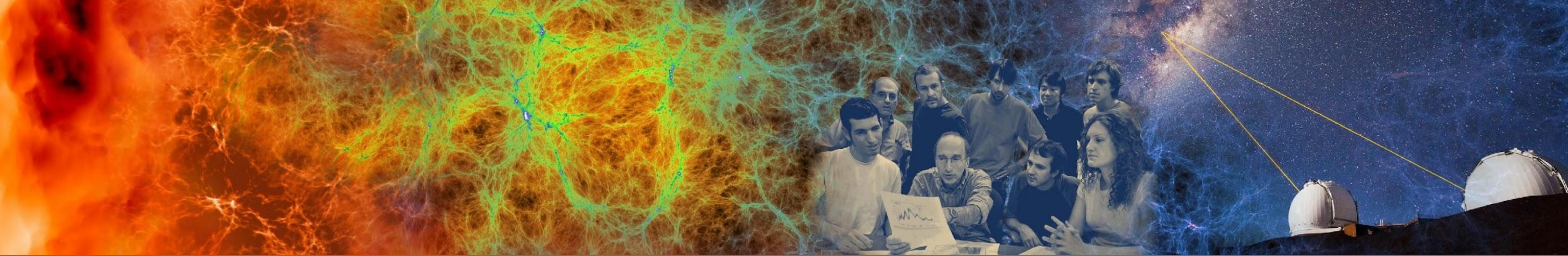
We expect an overlap to appear

Methodology

- Survey of public vendor roadmaps and literature on quantum resource estimates
- Prioritized unifying all data against a single metric

Caveats

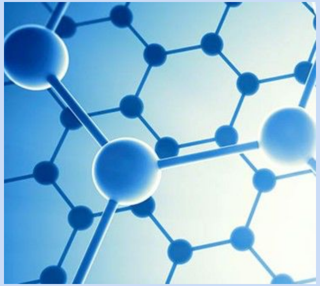
- Different assumptions made in roadmaps/estimates
- Variations in underlying hardware model
- Differences in accounting for overhead
- Leading order estimates
- Unknown unknowns
- Trade offs in implementations
- Classical hardness of problems
- Levels of optimism
- Based on public data only
- …



How large of a quantum computer will NERSC need?

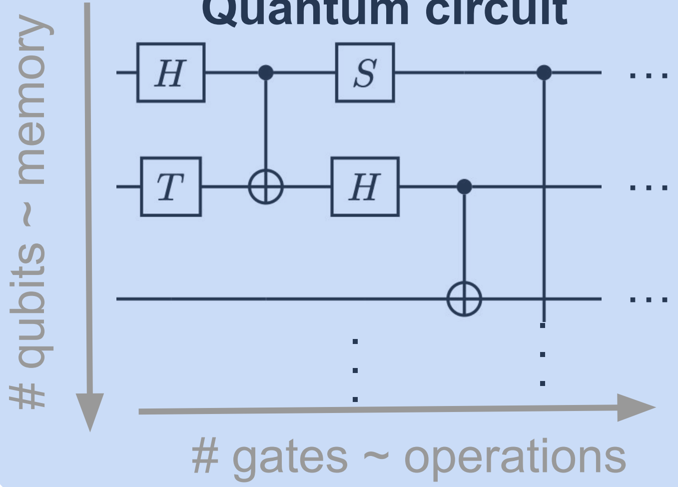
How do we measure quantum resources?

Domain problem



- problem encoding
- quantum algorithm
- state preparation
- pre/post-processing
- …

Quantum circuit

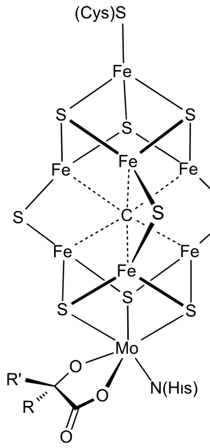


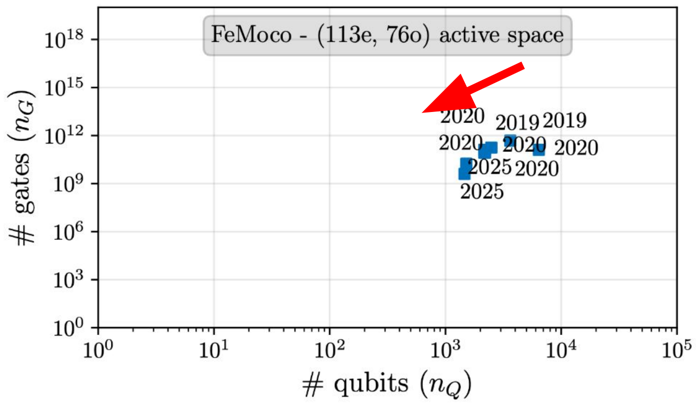
Quantum resource estimation

Unified space-time metric – `𝒫-vector': (# qubits, # gates)

We surveyed the literature and collected over 140 resource estimates.

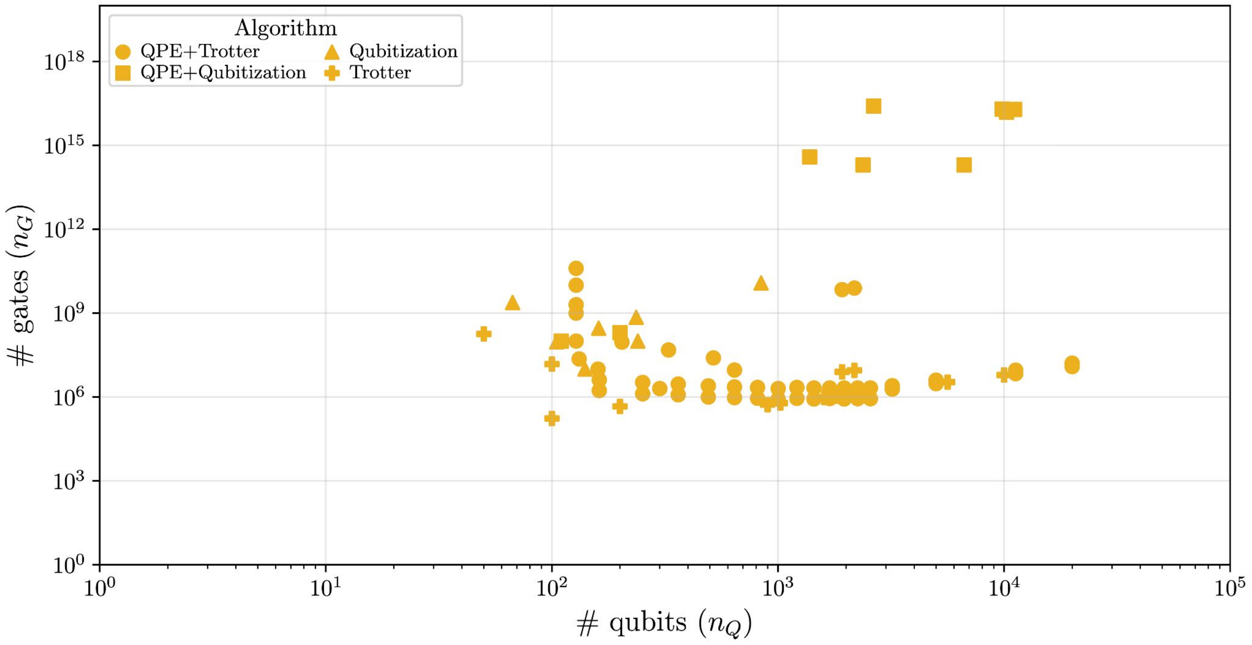
FeMoco benchmark – a chemistry case study





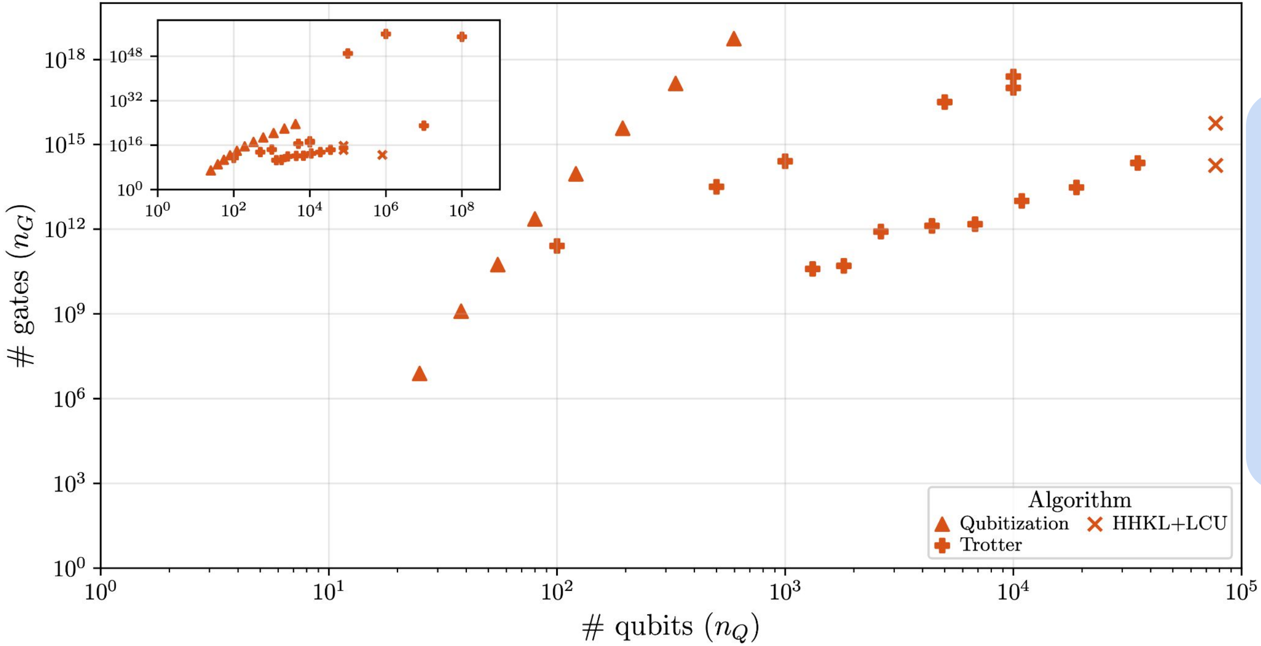
- Primary cofactor of nitrogenase – enzyme that catalyzes nitrogen fixation.
- Improved understanding of chemistry could help save few % of world's energy.
- Benchmark problem for quantum chemistry quantum algorithms.
- In the past 5 years, algorithmic improvements have reduced gate requirements by 1000x and qubit requirements by 5x.
- About 30 other resource estimates included in white paper

Condensed matter physics



- 80+ resource estimates
- Quantum dynamics & ground state energy estimation
- Problems include:
    - New cathode materials for Li-ion batteries
    - Exotic phases of matter

High energy physics



32 resource estimates for HEP, including:

- Heavy ion simulation
- Schwinger model
- Neutrino oscillations

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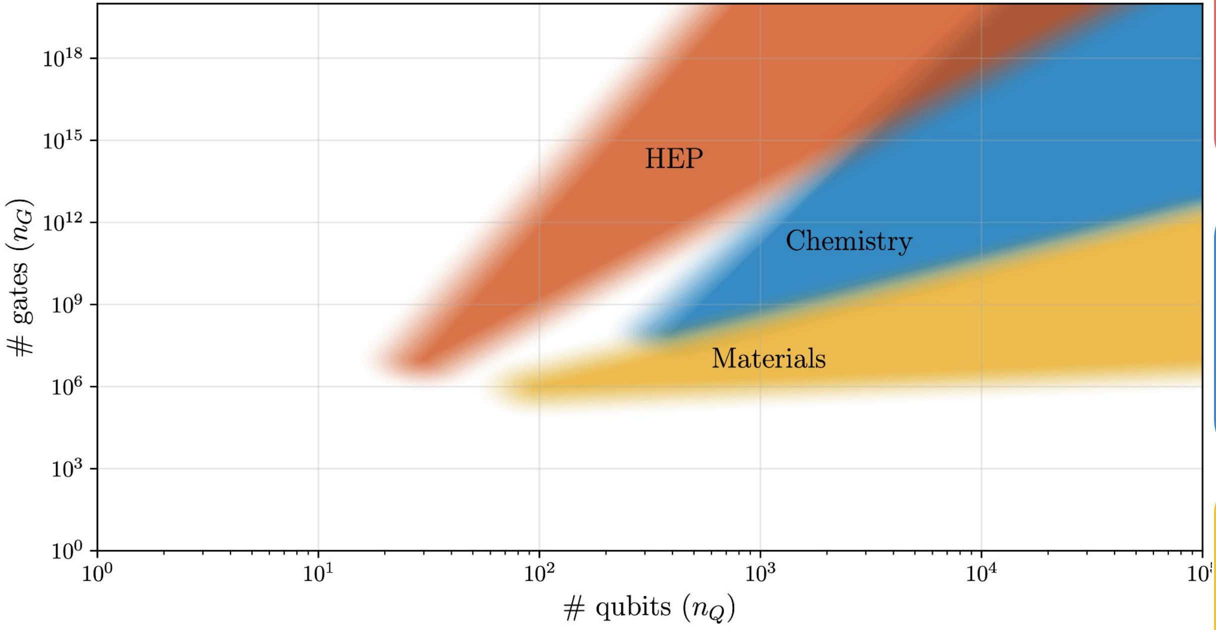
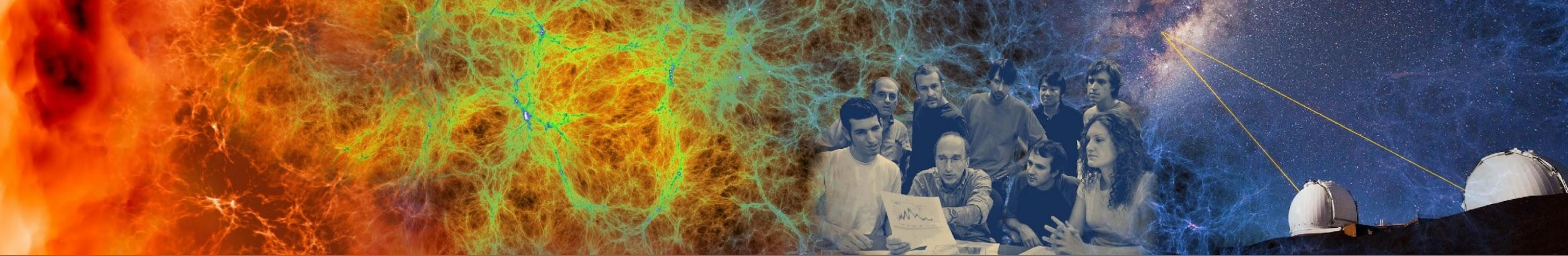
Overview of application requirements



HEP: 30+ estimates
- Largest number of gates
- Largest encoding overhead
- Algorithms less mature

Chemistry: 30+ estimates
- Intermediate number of gates
- Large encoding overhead
- Algorithms quite mature

Materials: 80+ estimates
- Fewest number of gates
- Lowest encoding overhead



How good are the quantum computers on the horizon?

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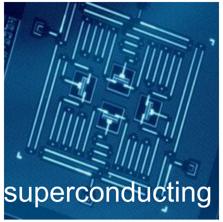
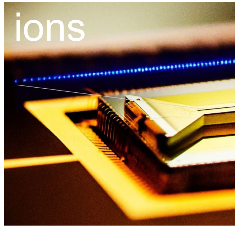
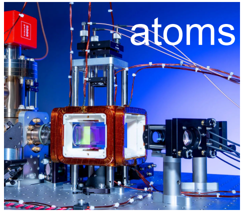
Our analysis of quantum vendor roadmaps

- Surveyed **public technology roadmaps** of quantum computing companies, including:
  - 6 superconducting
  - 2 trapped ion
  - 2 neutral atoms





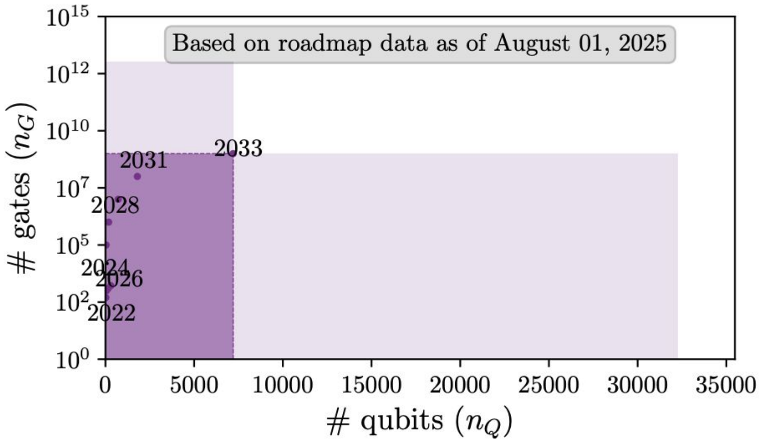


- **Unified milestone data** according to the $\mathscr{P}$-vector (#qubits, #gates) metric
- Estimated the effects of **quantum error mitigation** and **quantum error correction** on hardware performance

Consolidated every roadmap in a standard format

Year	Type	n_Q Phys.	n_Q Log.	error
2024	N	54	–	1×10^{-3}
2025	N	150	–	8×10^{-4}
2026	N	300	–	6×10^{-4}
2027	EF	10^3	36	10^{-5}
2028	EF	5×10^3	180	10^{-6}
2030	F	4×10^4	720	10^{-7}
2031	F	10^5	1,800	10^{-8}
2033	F	10^6	7,200	10^{-9}

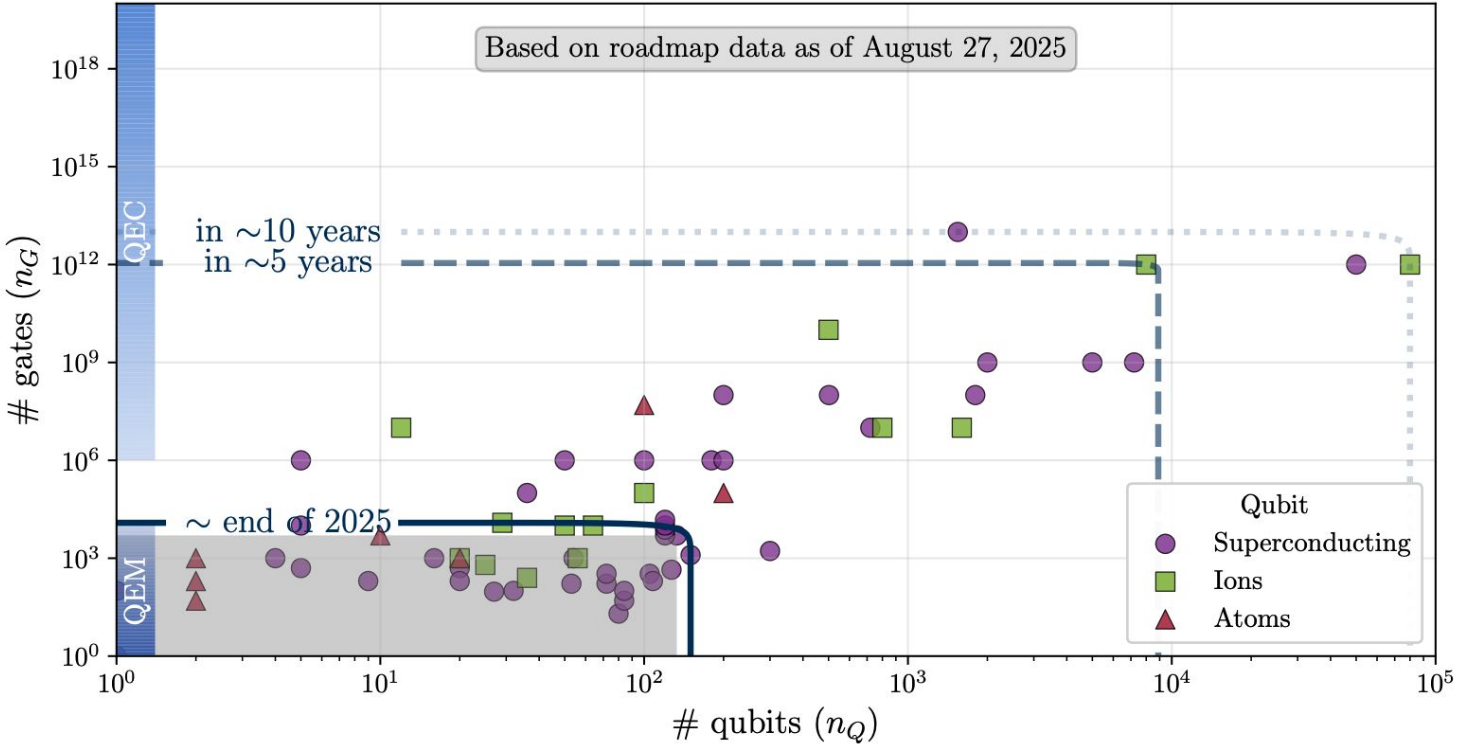
IQM Public roadmap



N: NISQ
EF: Early Fault-Tolerant
F: Fault-Tolerant

Error or other proxy for max depth

Overview of all roadmap milestones



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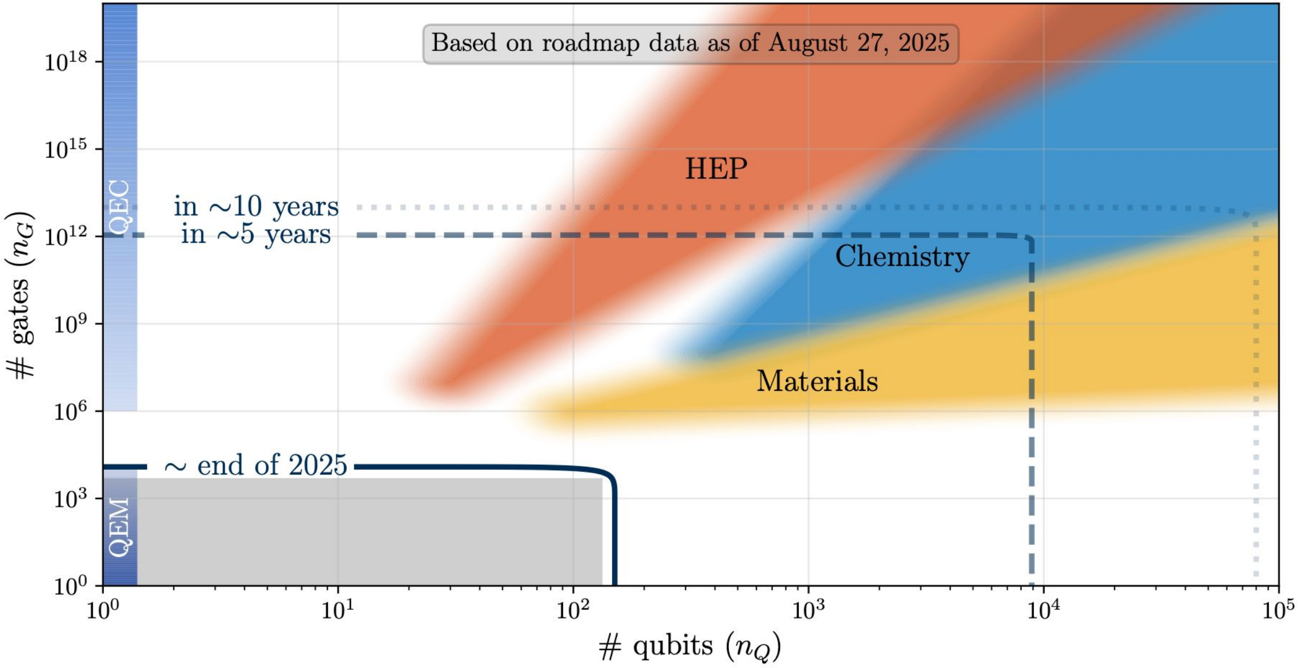

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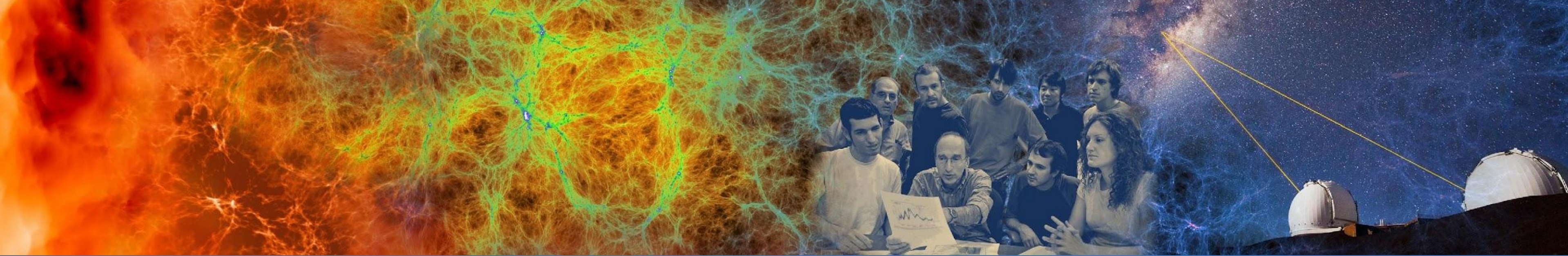

How fast do future quantum computers need to be?

Sustained quantum system performance (SQSP)

- Quantum analog of NERSC's SSP metric [Kramer, Shalf, Strohmaier 2005]
- Geometric mean of throughput for applications in benchmark suite
- We use logical clock frequency as proxy to estimate throughput*

System Scale	Clock Frequency	SQSP / Year
Megaquop	1 kHz	~300
	1 MHz	~300,000
	1 GHz	~300,000,000
Gigaquop	1 kHz	~2
	1 MHz	~2,000
	1 GHz	~2,000,000

* serialized resource state generation

Conclusion

- Quantum computing is making **rapid progress**: algorithms becoming more efficient, computers becoming better.
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